MIL-W-81044/6C
28 March 1986
SUPERSEDING
MIL-W-81044/6B
7 September 1976

MILITARY SPECIFICATION SHEET

WIRE, ELECTRIC, CROSSLINKED POLYALKENE INSULATED, TIN-COATED COPPER, NORMAL WEIGHT, 600-VOLT, 150°C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in effect of specification MIL-W-81044.

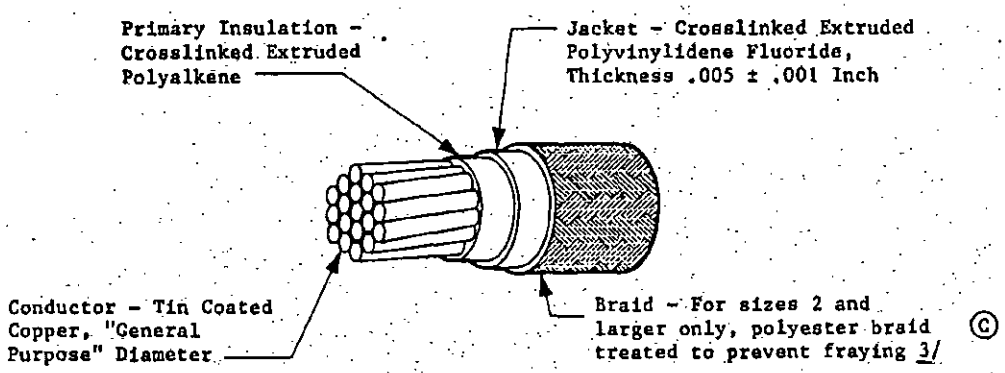


TABLE I. Construction details.

Part No. 1/	Wire size	Stranding (number of strands x AWG gage of strands)	Diameter of stranded conductor (inches)		Finished wire			
			(min)	(max)	Resistance at 20°C (68°F) (ohms/1000 ft) (max)	Diameter (inches)	Weight (lbs/1000 ft) (nom) 2/	Weight (lbs/1000 ft) (max)
M81044/6-24-*	24	19 x 36	.023	.026	26.2	.057 ± .002	2.8	3.0
M81044/6-22-*	22	19 x 34	.029	.033	16.2	.069 ± .003	4.1	4.3
M81044/6-20-*	20	19 x 32	.037	.041	9.88	.078 ± .003	5.8	6.1
M81044/6-18-*	18	19 x 30	.046	.051	6.23	.088 ± .003	8.3	8.7
M81044/6-16-*	16	19 x 29	.052	.058	4.81	.097 ± .004	10.3	10.9
M81044/6-14-*	14	19 x 27	.065	.073	3.06	.115 ± .004	15.5	16.2
M81044/6-12-*	12	37 x 28	.084	.090	2.02	.131 ± .004	23.0	24.2
M81044/6-10-*	10	37 x 26	.106	.114	1.26	.161 ± .005	35.3	36.5
M81044/6-8-*	8	133 x 29	.158	.173	.701	.219 ± .006	61.7	65.0
M81044/6-6-*	6	133 x 27	.198	.217	.445	.269 ± .007	94.5	103.
M81044/6-4-*	4	133 x 25	.250	.274	.280	.325 ± .008	150.	158.
M81044/6-2-*	2	665 x 30	.320	.340	.183	© .418 ± .013	232.	252.
© M81044/6-1-*	© 1	© 817 x 30	© .360	© .380	© .149	© .460 ± .015	281. ©	305. ©
M81044/6-0-*	0	1045 x 30	.405	.425	.116	© .508 ± .017	352.	380.

1/ Part No.: The asterisks in the part number column, Tables I and II, shall be replaced by color code designators in accordance with MIL-STD-681. Examples: Size 20, white - M81044/6-20-9; white with orange stripe - M81044/6-20-93.

2/ Nominal values are for information only. Nominal values are not requirements.

© 3/ Braid (sizes 2 and larger): Bright polyethylene terephthalate polyester yarn, 220 denier, 34 filaments minimum, braided 4 ends per carrier; braid to be tightly formed, uniform in appearance, and treated with a clear finisher coating. The finisher coating shall be compatible with the temperature rating and performance requirements of the insulated wire. Braid colors shall be limited to solid brown, solid red and solid white.

© Denotes changes.

AMSC N/A FSC 6145

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TABLE II. Performance details. Ⓒ

Part no.	Bend testing				
	Mandrel diameter (in.) (±3%)			Test load (lbs) (±3%)	
	Life cycle, accelerated aging and immersion	Cold bend	Wrap	Life cycle, accelerated aging and immersion	Cold bend
M81044/6-24-*	.50	1.0	.250	1.0	3.0
M81044/6-22-*	.75	1.0	.250	3.0	3.0
M81044/6-20-*	.75	1.0	.250	4.0	4.0
M81044/6-18-*	1.00	1.5	.375	4.0	4.0
M81044/6-16-*	1.00	1.5	.375	5.0	5.0
M81044/6-14-*	1.50	2.0	.50	5.0	5.0
M81044/6-12-*	2.0	2.0	.50	5.0	5.0
M81044/6-10-*	3.0	3.0	.75	5.0	5.0
M81044/6-8-*	3.0	4.0	.75	6.0	6.0
M81044/6-6-*	4.0	5.0	1.00	6.0	10.0
M81044/6-4-*	5.0	6.0	1.25	6.0	10.0
M81044/6-2-*	6.0	8.0	3.0	6.0	15.0
Ⓒ M81044/6-1-*	Ⓒ 8.0	Ⓒ 10.0	Ⓒ 4.0	Ⓒ 7.0	Ⓒ 15.0
M81044/6-0-*	8.0	10.0	4.0	7.0	15.0

Wire ratings and additional requirements

Temperature rating: 150°C (302°F) max conductor temperature
Voltage rating: 600 volts (rms) at sea level
Ⓒ Accelerated aging: Oven temperature, 300° ± 2°C (572° ± 3.6°F) for 6 hours (Sizes 2 and larger, test prior to application of braid.) For identification legibility, 225° ± 2°C (437° ± 3.6°F) for 6 hours, test not required for sizes 2 and larger
Blocking: 225° ± 2°C (437° ± 3.6°F)
Ⓒ Color: Sizes 24 through 4: In accordance with MIL-STD-104, Class 1; white preferred
Sizes 2 through 0: Braid color shall be in accordance with MIL-STD-104, Class 2, except that brown shall be within Munsell color limits of 5YR 5/4 and 5R 3/6; nominal, 2.5YR 3/6. Jacket color under braid shall be white, MIL-STD-104, Class 1.
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper
Humidity resistance: 5000 megohms for 1000 ft, min insulation resistance after humidity exposure
Ⓒ Identification, striping or banding durability (not required for sizes 2 and larger): 125 cycles (250 strokes) (min), 500 grams weight
Impulse dielectric test:
Primary insulation (when test is used in lieu of spark test): 6.0 kilovolts (peak), 100% test
Finished wire: 8.0 kilovolts (peak), 100% test
Insulation resistance: 5000 megohms for 1000 ft (min)
Ⓒ Life cycle: Oven temperature 200° ± 2°C (392° ± 3.6°F) for 168 hours (Sizes 2 and larger, test prior to application of braid.)

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Physical properties of primary insulation:
Tensile strength, 2500 psi (min) for sizes 24 through 12
2000 psi (min) for sizes 10 through 0
Elongation, 150% (min) for all sizes

© Shrinkage:
Sizes 24 through 12: 0.125 inch max at 300° ± 2°C (572° ± 3.6°F)
Sizes 10 through 0: 0.250 inch max at 300° ± 2°C (572° ± 3.6°F)
(Sizes 2 and larger, test prior to application of braid.)

Smoke: 200° ± 2°C (392° ± 3.6°F)

Spark test of primary insulation: 3000 volts (rms), 60 Hz, 100% test

© Surface resistance:
500 megohms-inches (min), initial and final readings (Sizes 2 and larger, test prior to application of braid.)

© Thermal shock: Oven temperature 150° ± 2°C (302° ± 3.6°F)
Maximum change in measurement: 0.060 inch except no requirement for braid layer on sizes 2 and larger

Wet dielectric test: 2500 volts (rms)

Wicking: 2.25 inches (max) between primary insulation and jacket

© Wrap test: Mandrel wrap test required (Sizes 2 and larger, test prior to application of braid.)

© Metric conversion note: Data in this specification may be converted to metric as follows:

Linear dimensions	25.40 x inches = millimeters (mm)
Weight (general)	.4536 x lbs = kilograms (kg)
Wire weight	1.488 x (lbs/1000 ft) = kg/km
Conductor resistance	3.281 x (ohms/1000 ft) = ohms/km
Insulation resistance	.3048 x (megohms for 1000 ft) = megohms for 1 km
Surface resistance	Metric documents may differ from MIL-STD-228 in spacing of the electrodes for this determination and in the method of expressing the test results. Where the electrode spacing is 25.0 millimeters in the metric document, the MIL-W-81044 megohm-inches resistance (defined as total megohms resistance times inches wire diameter) may be converted as follows:

25.0 x (megohm-inches diameter) = megohm-mm diameter
78.5 x (megohm-inches diameter) = megohm-mm circumference
3.14 x (megohm-inches diameter) = megohm-mm circumference per mm of electrode spacing

Custodians:
Navy - AS
Army - CR
Air Force - 85

User activities:
Navy - MC, OS

Review activities:
Navy - EC
Army - AT, AV, MI
Air Force - 99
NSA - NS
DLA - IS

Preparing activity:
Navy - AS
(Project No. 6145-1003)

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